

3.1 Tectonic Hazards

Geography — Eduqas A-Level (A110QS) — Component 3: Contemporary Themes

1. Plate Tectonics Theory — The Unifying Framework

Plate tectonics is the unifying theory of modern solid-Earth geoscience and the physical foundation of this topic. Earth's outer shell, the **lithosphere** (crust plus uppermost rigid mantle, ~100 km thick under continents, thinner under oceans), is broken into **seven major plates**, around eight significant minor plates and many smaller microplates. These rigid plates "float" on the partially molten, ductile **asthenosphere** beneath, moving at rates of 1–15 cm per year — roughly the rate at which human fingernails grow. All earthquake, volcanic and tsunami activity is ultimately explained by the relative motion of these plates and by mantle plumes beneath them.

Plate tectonics: The theory that Earth's lithosphere is divided into rigid plates whose relative motion over the ductile asthenosphere generates earthquakes, volcanoes, mountain belts and the geometry of ocean basins. Driven by mantle convection, slab pull and ridge push.

Lithosphere: The rigid outer shell of Earth comprising the crust and the uppermost rigid portion of the mantle. Typically ~100 km thick under continents and ~70 km under oceans. Behaves elastically on short timescales and so can store and release seismic strain.

Asthenosphere: The weak, partially molten layer of the upper mantle below the lithosphere (approximately 100–350 km depth). Flows plastically on geological timescales, permitting plate motion.

1.1 Historical development of the theory

The modern theory emerged in stages across the twentieth century:

- **Alfred Wegener (1912)** — German meteorologist who proposed *continental drift*: the idea that Earth's continents were once united in a supercontinent ("Pangaea") that fragmented and drifted apart. Evidence: the fit of the Atlantic coastlines of Africa and South America; identical *Mesosaurus* fossils on both sides of the South Atlantic; glacial striations matching across continents; coal seams (tropical origin) now in Antarctica. Rejected by most contemporaries because Wegener could not identify a plausible mechanism.
- **Arthur Holmes (1929)** — proposed mantle convection as a possible driver.
- **Paleomagnetic work (1950s)** — Runcorn and others showed that the apparent position of the magnetic pole had "wandered" differently for Europe and North America — consistent not with true polar wander but with the two continents having moved relative to each other.
- **Harry Hess (1962) — sea-floor spreading.** New oceanic crust is generated at mid-ocean ridges and moves outwards, being ultimately destroyed at subduction zones. The ocean floor is a conveyor belt.
- **Vine and Matthews (1963)** — symmetrical magnetic "stripes" either side of the Mid-Atlantic Ridge record geomagnetic reversals, providing decisive confirmation of sea-floor spreading.
- **GPS geodesy (1990s–present)** — centimetre-scale measurements of plate velocity now confirm the model in real time. The Hawaiian Islands, for example, drift north-west at ~7 cm/yr relative to the mantle plume beneath them.

Exam tip — evidence for plate tectonics: When asked about evidence, move beyond Wegener's jigsaw-fit. The *decisive* evidence is paleomagnetic striping (Vine & Matthews), age of ocean-floor sediments (youngest at ridges), symmetrical heat-flow patterns, and modern GPS. Examiners reward candidates who know that continental drift was only *hypothesised* in 1912 but *confirmed* by ocean-floor evidence half a century later.

1.2 Driving mechanisms

Plate motion is driven by three linked processes. **Mantle convection** — the slow rise of hot, buoyant material beneath ridges and its sinking at subduction zones — was the classical explanation and is still the ultimate energy source. Current consensus gives more weight to two "edge forces":

- **Slab pull:** as cold, dense oceanic lithosphere subducts, it pulls the rest of the plate after it. Generally considered the dominant force — plates with long subducting edges (e.g. Pacific, Nazca) move faster.

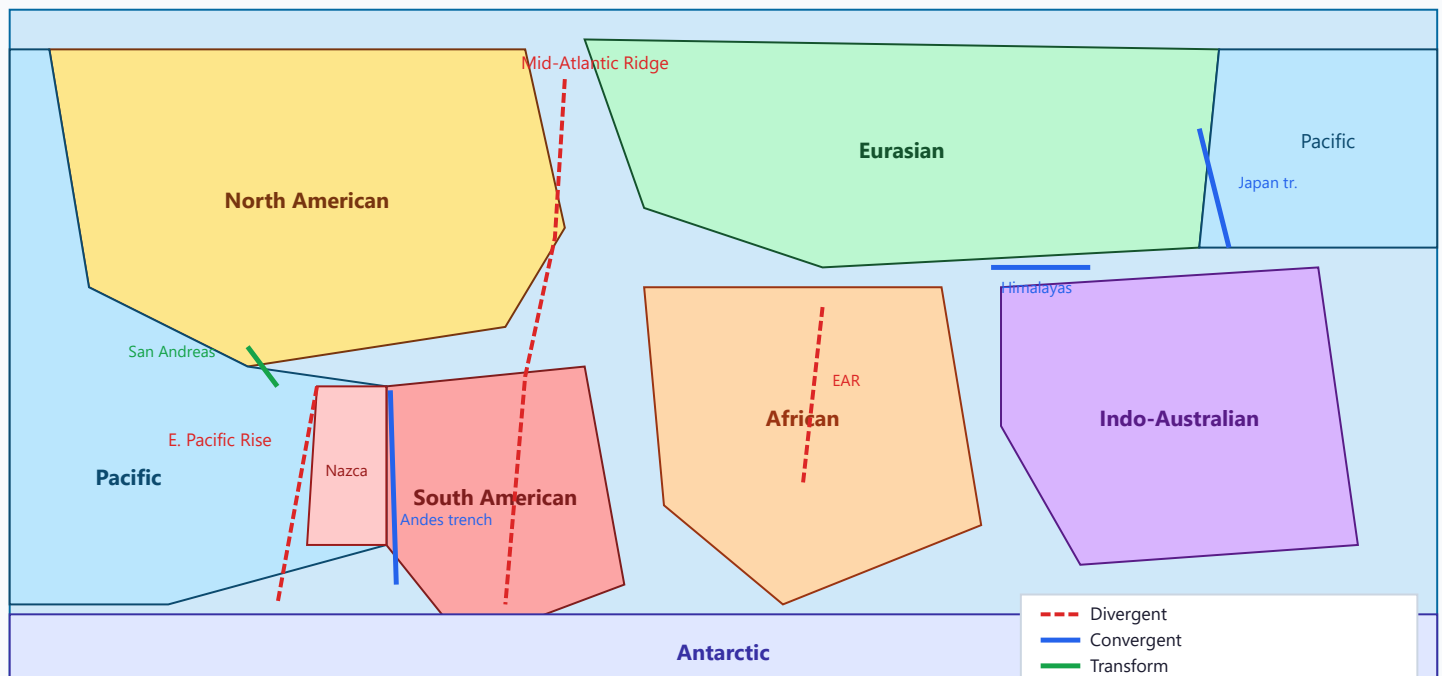
- **Ridge push:** elevated mid-ocean ridges gravitationally slide their flanks outwards. Secondary in magnitude but ubiquitous.
- **Basal drag:** friction between lithosphere and asthenosphere can either drive or retard motion depending on relative velocities.

Slab pull: The gravitational force generated by a cold, dense slab of oceanic lithosphere sinking at a subduction zone, which pulls the rest of the plate towards the trench. The dominant driver of plate motion in current models.

2. The Global Plate Map

Knowing the plate map at a recognitional level is essential: case studies always need to be located with reference to named plates and boundaries. The seven major plates are: **Pacific, North American, South American, Eurasian, African, Antarctic, Indo-Australian**. Significant minor plates include the **Nazca, Cocos, Caribbean, Philippine, Arabian, Juan de Fuca, Scotia** and **Caroline** plates.

Major lithospheric plates — schematic world map



3. Plate Boundary Types

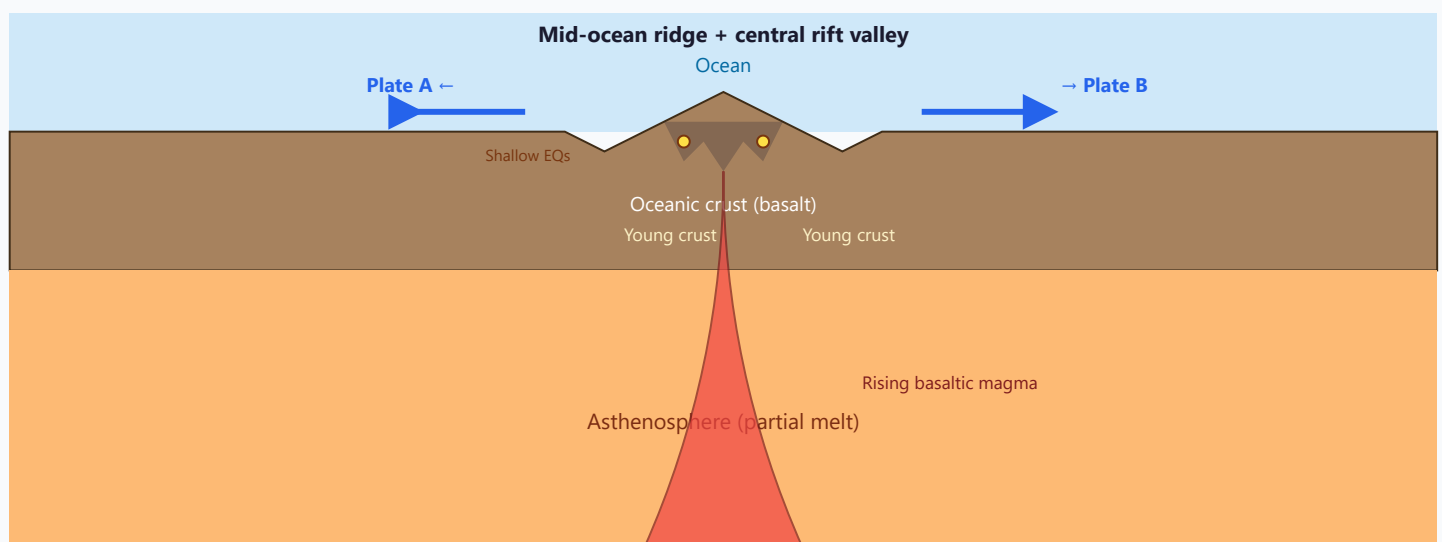
Three kinematic classes of plate boundary are recognised, each producing a characteristic suite of landforms and hazards. A fourth hazard locus — the *hot spot* — sits independently of plate boundaries.

3.1 Divergent (constructive) boundaries

Two plates move apart. New oceanic lithosphere is generated as basaltic magma rises from the asthenosphere and solidifies. The characteristic landform is a **mid-ocean ridge**, such as the Mid-Atlantic Ridge (spreading at ~2.5 cm/yr), the East Pacific Rise (a much faster ~15 cm/yr), and the Carlsberg Ridge in the Indian Ocean. Where divergence occurs *within a continent*, the result is a **rift valley**: the East African Rift (from Mozambique through Ethiopia to the Red Sea) is a classic example, producing volcanoes such as Kilimanjaro, Nyiragongo and the Afar lava lakes. Iceland sits astride the Mid-Atlantic Ridge *and* a hot spot, which is why it is so volcanically active.

Hazards at divergent boundaries tend to be *less catastrophic* than at convergent boundaries: volcanism is typically *effusive* (fluid basalt flows such as the 2014 Holuhraun eruption, which emitted more lava than any Icelandic event since Laki 1783 but caused no deaths), and earthquakes are generally *shallow and modest* ($M_w < 6.5$) because the crust is thin and the rocks are hot and ductile.

Divergent boundary — mid-ocean ridge (cross-section)



3.2 Convergent (destructive) boundaries

Two plates move towards each other. The outcome depends on the *type of lithosphere* involved:

Oceanic–continental convergence

The denser oceanic plate *subducts* beneath the less dense continental plate, producing: a deep **ocean trench** (e.g. Peru–Chile Trench, 8 km deep); a **Benioff–Wadati zone** of earthquakes dipping from shallow

to very deep under the continent; partial melting of the subducted slab and overlying mantle wedge, producing *andesitic to dacitic* magmas which feed explosive **stratovolcanoes** arranged along a continental **volcanic arc** (the Andes are the textbook case; the Cascades of Washington and Oregon are similar). Crustal shortening produces fold-and-thrust mountain belts paralleling the arc.

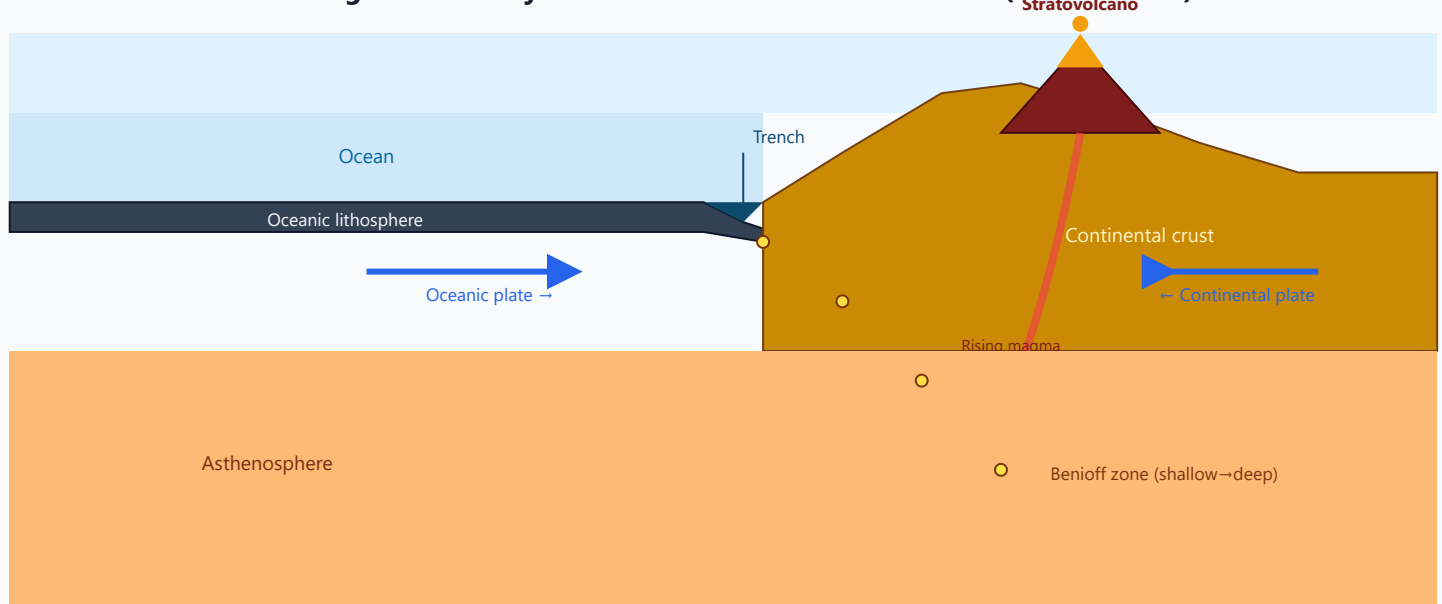
Oceanic–oceanic convergence

The older, denser oceanic plate subducts beneath the younger one, producing an **island arc** (an arcuate chain of volcanic islands) paralleling a trench. Examples: the Japanese, Philippine, Mariana, Kurile, Tonga, Aleutian and Lesser Antilles arcs. The *deepest* points on Earth's surface lie in these trenches — the Mariana Trench reaches ~11 km.

Continental–continental convergence

Neither plate is dense enough to subduct. Instead the two buoyant continental masses *collide*, producing intense *uplift* but typically *no volcanism* (because no wet oceanic slab is descending into the mantle to drive melting). Earthquakes are shallow to intermediate and can be very large. The Himalayas (where the Indian plate is colliding with the Eurasian plate at ~4–5 cm/yr, producing Everest and Kangchenjunga) and the Alps (African plate against Eurasian) are classic examples. Tibet has been elevated ~5 km by this process and thickened to ~70–80 km crust.

Convergent boundary — oceanic–continental subduction (cross-section)

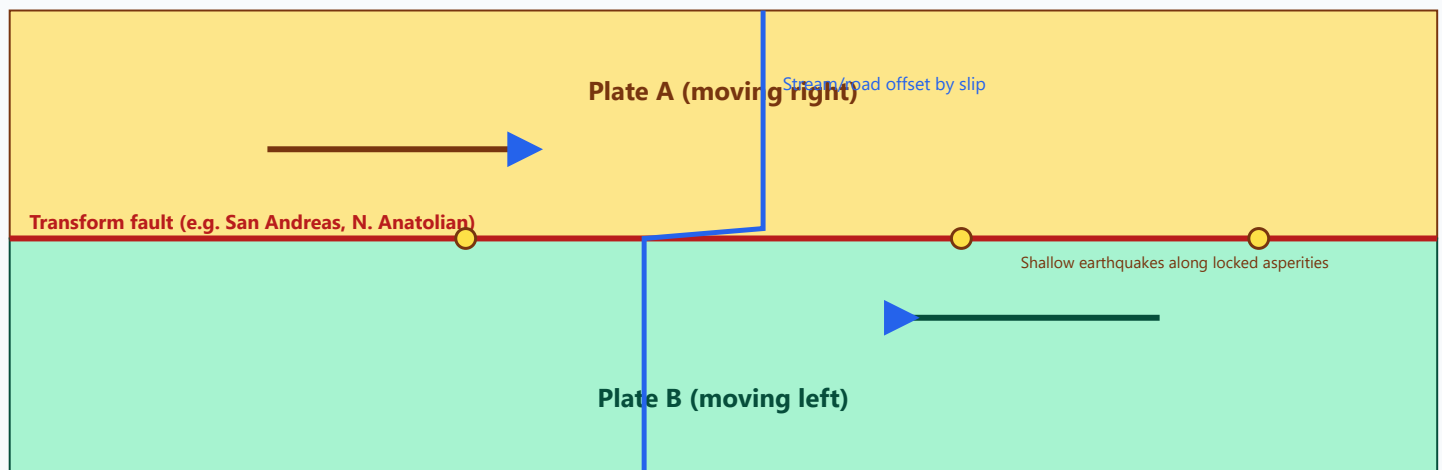


3.3 Transform (conservative) boundaries

Plates slide horizontally past each other. No crust is created or destroyed, and there is no volcanism. Stress accumulates until it is released as an earthquake — often a very large one because faults can be locked for

centuries before rupturing. The **San Andreas Fault** in California is the most famous example, marking the boundary where the Pacific Plate moves north-westwards past the North American Plate at ~3–5 cm/yr. The **North Anatolian Fault** in Turkey has produced a famous westward-migrating sequence of earthquakes since 1939. The **Alpine Fault** of New Zealand's South Island is another example. Transform boundaries on land tend to cross densely populated regions, which makes them disproportionately hazardous relative to their plate-tectonic importance.

Transform boundary — plan view with displacement



3.4 Hot spots

Narrow, deep-mantle *plumes* of hot material generate volcanism that is largely *independent* of plate boundaries. As a plate drifts across a stationary plume, a **chronological chain of volcanoes** is produced, youngest above the plume. The Hawaiian–Emperor chain is the textbook case: the Big Island (<0.5 Ma) sits on the plume, Kauai is ~5 Ma, and the Emperor seamounts extending north-westwards record motion over 80 Ma. Iceland, Yellowstone, La Réunion and the Galápagos are other hot spots; when a hot spot is coincident with a divergent boundary (Iceland) its output is especially large.

Hot spot: A long-lived localised zone of volcanism believed to result from a deep-mantle plume of hot material that melts through the overriding lithosphere. Produces age-progressive volcanic chains as plates drift across the plume.

Exam tip — boundary type determines hazard profile: Always specify the boundary type when describing a tectonic setting. Convergent destructive zones produce the most varied and severe hazards (deep earthquakes, explosive volcanoes, tsunamis, lahars); transform boundaries produce

large shallow earthquakes but no volcanism; divergent boundaries produce effusive volcanism and moderate earthquakes. A one-sentence boundary classification early in an answer anchors everything that follows.

3.5 Summary table of boundary types

Boundary type	Relative motion	Key landforms	Earthquake character	Volcanism	Named example
Divergent (oceanic)	Plates move apart	Mid-ocean ridge, transform-offset ridge segments	Shallow, moderate ($M_w < 6.5$)	Effusive basalt	Mid-Atlantic Ridge
Divergent (continental)	Plates move apart	Rift valley, block-faulted basins	Shallow, moderate	Mixed shield + explosive	East African Rift
Convergent ocean–continent	Oceanic subducts	Trench, volcanic arc, fold mountains	Shallow to very deep; mega-thrusts possible	Explosive andesite/dacite	Andes, Cascades
Convergent ocean–ocean	Older oceanic subducts	Trench, island arc	Shallow to very deep; mega-thrusts	Explosive	Japan, Mariana, Philippines
Convergent continent–continent	Collision	Fold mountains, plateau uplift	Shallow to intermediate; very large	Typically none	Himalayas, Alps
Transform	Plates slide past	Linear fault trace, offset features	Shallow, often very large	None	San Andreas, N. Anatolian
Hot spot	Independent of plates	Age-progressive	Minor, localised	Basaltic shield (usually)	Hawaii, Iceland,

4. Earthquakes — Processes and Measurement

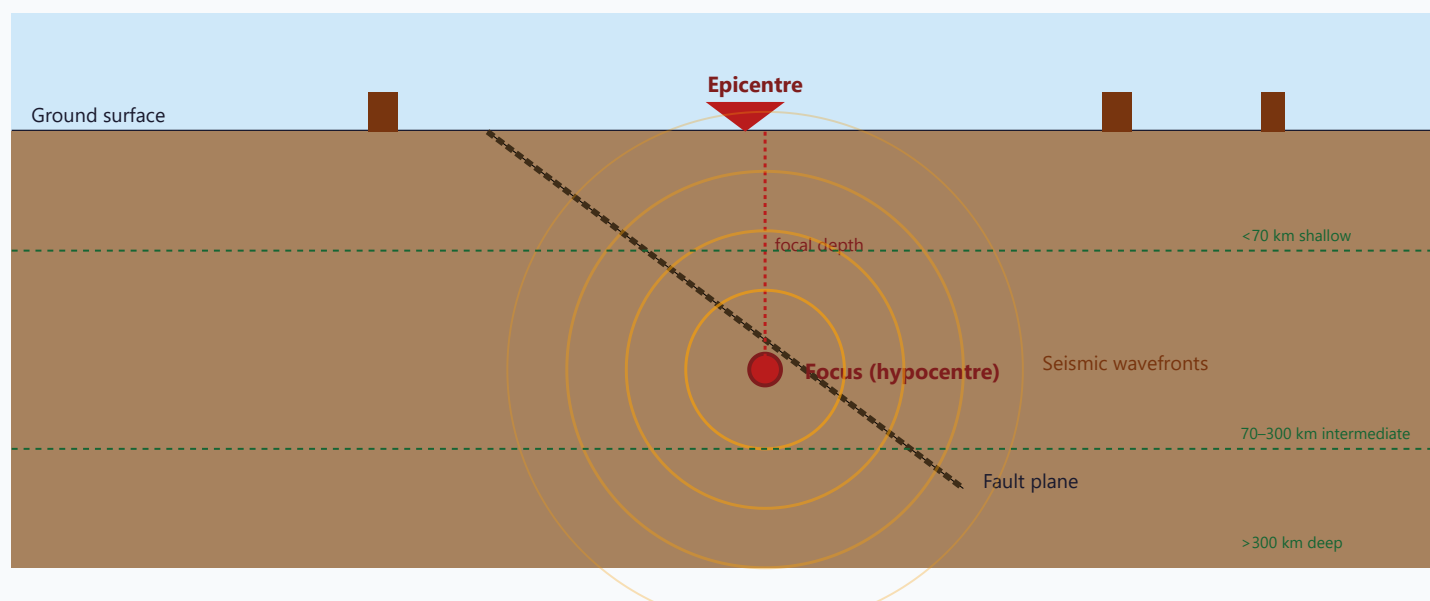
Earthquakes are the sudden release of elastic strain energy accumulated along faults as a result of plate motion. The theory was first formalised by H. F. Reid after the 1906 San Francisco earthquake as the *elastic rebound theory*.

4.1 Focus, epicentre and rupture

Focus (hypocentre): The point within the Earth, on the fault plane, where rupture *begins* and seismic waves radiate outwards. Measured as a depth below the surface.

Epicentre: The point on the Earth's surface *directly above* the focus. Intensity of ground shaking is typically greatest here, though the directivity of rupture and site effects (soft sediments, topography) can displace the worst damage.

Earthquake focus, epicentre and wave propagation



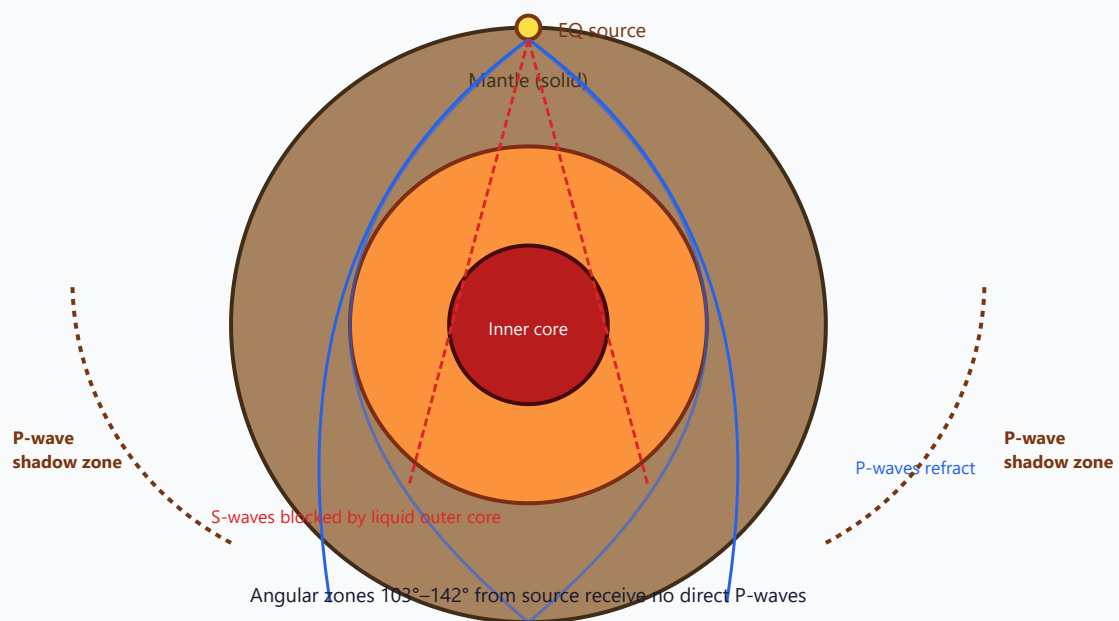
4.2 Seismic waves

The rupture radiates several kinds of elastic wave, each useful to seismologists:

- **P-waves (primary, compressional):** fastest (~6–8 km/s in the crust); particles oscillate *parallel* to the direction of travel (push–pull); able to travel through *both solids and liquids*; arrive first at a seismograph.
- **S-waves (secondary, shear):** slower (~3.5–4.5 km/s); particles oscillate *perpendicular* to travel direction; *cannot* pass through liquids — this property allowed seismologists (notably Harold Jeffreys, 1926) to infer a liquid outer core; Inge Lehmann (1936) later used P-waves to discover the solid inner core.
- **Surface waves** — produced when body waves reach the surface:
 - *Love waves*: horizontal shearing; destructive to foundations.
 - *Rayleigh waves*: rolling, elliptical motion like ocean waves; cause most of the perceived ground roll.

Because P-waves travel faster than S-waves, the time lag between their arrivals at a seismograph is a direct measure of distance from the epicentre — three stations are enough to triangulate the epicentre. This principle is the basis of modern **early-warning systems** (Section 9).

P- and S-wave shadow zones through the Earth



4.3 Magnitude and intensity scales

Three scales are important — candidates often confuse magnitude (an intrinsic property of the earthquake) with intensity (how it is felt at a place).

Richter magnitude (M_L): The original local-magnitude scale devised by Charles Richter in 1935 for southern California. Based on the maximum amplitude on a Wood–Anderson seismograph, corrected for distance. Logarithmic — each unit is $10\times$ the amplitude and roughly $32\times$ the energy. Saturates above $\sim M7$, so superseded for large events.

Moment magnitude (M_w): Current global standard, based on seismic moment ($M_0 = \text{rigidity} \times \text{rupture area} \times \text{slip}$). Captures the total energy released, does not saturate at high values, and is therefore the only scale that can meaningfully describe events above $M8$. Each unit increase = $\sim 32\times$ more energy released.

Mercalli intensity scale (Modified Mercalli Intensity, MMI): An ordinal 12-point scale describing the *observed effects* at a particular location — from I (not felt) to XII (total destruction). Depends on distance from epicentre, depth, local geology and building stock. The same earthquake has a single magnitude but many MMI values.

Energy in context

Because the scale is logarithmic, differences of one unit are very large:

- An $M6$ event releases approximately the energy of the Hiroshima atomic bomb (~ 15 kt TNT).
- An $M7$ event releases $\sim 32\times$ more — roughly 500 kt, comparable to a modern thermonuclear weapon.
- An $M9$ event (e.g. Tōhoku 2011) releases $\sim 1,000\times$ more energy than an $M7$ — roughly the world's total energy consumption for a day, released in a few minutes.

4.4 Magnitude scale comparison

Scale	Basis	Type	Range used	Limitations
Richter (M_L)	Amplitude on Wood–Anderson seismograph	Logarithmic magnitude	$\sim M3$ – $M7$ (California)	Saturates above $\sim M7$; local only

Surface-wave (M_s)	Rayleigh-wave amplitude at 20 s period	Logarithmic magnitude	M5–M8	Saturates at very large events; only for shallow quakes
Body-wave (m_b)	P-wave amplitude	Logarithmic magnitude	Any depth	Saturates > M6.5; used for nuclear-test monitoring
Moment (M_w)	Seismic moment M_0	Logarithmic magnitude	Any size; current global standard	Requires full waveform modelling
Mercalli (MMI)	Observed damage and perception	Ordinal intensity (I–XII)	Everywhere felt	Subjective; depends on site, buildings

4.5 Depth categories

- **Shallow (0–70 km):** account for ~75% of earthquake energy release; generally the most damaging to surface infrastructure because the source is close. Almost all transform-boundary and rift-valley earthquakes are shallow.
- **Intermediate (70–300 km):** typical of deeper parts of subducting slabs.
- **Deep (>300 km, to ~700 km maximum):** always associated with subducting oceanic lithosphere (the Benioff–Wadati zone). A deep event of a given magnitude tends to produce *less* surface damage than a shallow event of the same magnitude because the seismic waves attenuate over a longer path.

Exam tip — always cite magnitude AND depth: A M_w 7.0 at 13 km depth (Haiti 2010) is enormously more damaging than a M_w 7.0 at 200 km depth (a typical deep Mariana event). Quoting magnitude alone is therefore incomplete. Also make sure you use M_w and not “the Richter scale” for any event above ~M7, which covers essentially every major case study.

5. Secondary Earthquake Hazards

5.1 Tsunami

A **tsunami** is a series of ocean waves generated when a large volume of water is suddenly displaced — most commonly by undersea fault rupture (a reverse-fault mega-thrust), but also by submarine landslides,

volcanic flank collapse (Anak Krakatoa 2018), or, rarely, asteroid impact. Tsunamis have very long wavelengths (often >100 km in the open ocean) and travel across deep ocean basins at speeds comparable to a jet airliner (~700–800 km/h). In the open ocean their amplitude is small (often <1 m) and they pass unnoticed beneath ships; as they enter shallow water, the wavelength shortens and the amplitude increases sharply (shoaling), producing devastating run-up on land — typically 5–15 m, exceptionally 40 m (Tōhoku 2011) or >80 m for landslide-generated events (Lituya Bay 1958, Alaska).

Because tsunamis travel at known speeds across known bathymetry, **warning is technically feasible** for basin-crossing events: the Pacific Tsunami Warning Center (Hawaii, since 1949) and the Indian Ocean Tsunami Warning System (established after 2004) use a network of DART (Deep-ocean Assessment and Reporting of Tsunamis) buoys that detect passing waves and relay data via satellite. For near-source events (rupture within ~50 km of the coast) warning times may be only a few minutes — exactly the problem at Tōhoku 2011 and Sumatra 2004.

5.2 Liquefaction

Liquefaction: The process by which water-saturated, loosely packed granular sediment (sand, silt) temporarily loses shear strength during seismic shaking and behaves as a viscous liquid. Buildings tilt, sink or topple; buried structures (tanks, pipes) can float to the surface; surface sediment vents as “sand boils.”

Particularly severe in reclaimed land, river deltas, and coastal sediments. Historically documented cases: Niigata 1964 (apartment blocks toppled onto their sides but remained intact); San Francisco Marina District 1989 (Loma Prieta); and the 2011 Christchurch (New Zealand) earthquake, where several km² of eastern suburbs were rendered uninhabitable and the government, through the Canterbury Earthquake Recovery Authority, offered to buy out entire neighbourhoods (“red zones”).

5.3 Landslides and slope failure

Seismic shaking can trigger mass movements on already unstable slopes. The 1970 Ancash earthquake (Peru, Mw 7.9) detonated a rock-and-ice avalanche from Huascarán that buried the town of Yungay and killed some 20,000 people in minutes. The 2018 Lombok (Indonesia) event produced widespread slope failures on the flanks of Mount Rinjani. Mountain-front subduction-zone earthquakes (Chile 1960, Alaska 1964, Tōhoku 2011) are particularly prone to this cascading hazard.

5.4 Building collapse and fire

Building collapse is the **primary cause of earthquake fatalities worldwide**, and is overwhelmingly a function of *construction quality* rather than magnitude. Unreinforced masonry, heavy concrete floors on slender columns, unbraced soft-storey ground floors and pancake collapse modes are the deadliest. Fire was the dominant secondary hazard of 19th- and early-20th-century earthquakes in wooden cities: the 1906 San Francisco quake killed perhaps 3,000 people largely because ruptured gas mains and broken water supplies left the fire department helpless for three days; the 1923 Kantō (Tokyo–Yokohama) earthquake on the Sagami Trough killed ~140,000, with the majority dying in the ensuing firestorm because the quake hit at lunchtime when charcoal stoves were lit.

6. Volcanic Processes and Types

6.1 Magma composition controls eruption style

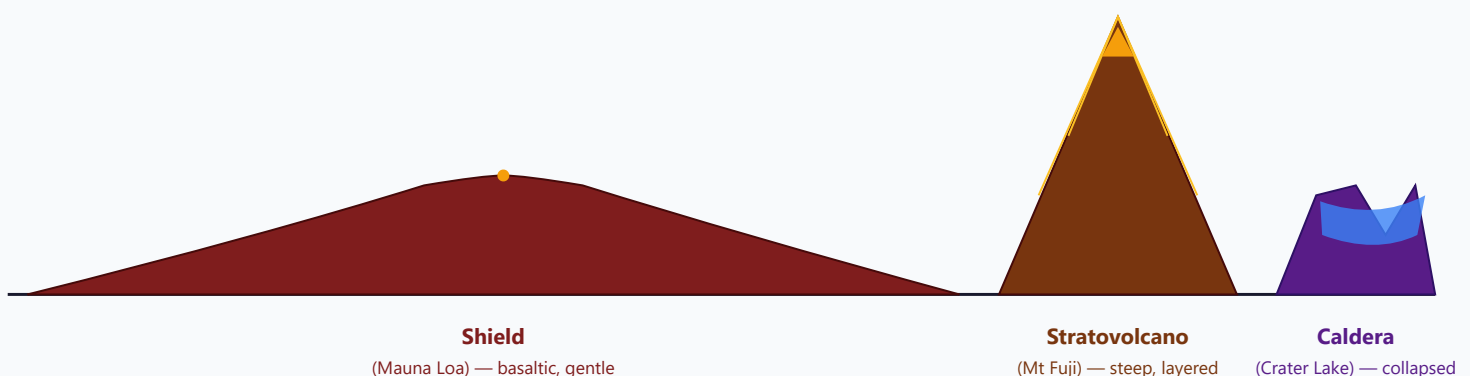
The single most important control on how a volcano erupts is the *chemistry* of its magma, which in turn depends on its tectonic setting. Silica (SiO_2) content and dissolved gas content together determine *viscosity*. Low-viscosity magma lets gas bubbles escape easily, producing effusive eruptions; high-viscosity magma traps gas until sudden catastrophic release produces explosive eruptions.

Magma type	SiO_2 (%)	Viscosity	Temperature (°C)	Typical setting	Eruption style
Basaltic (mafic)	45–52	Low (runny)	1100–1200	Hot spots, divergent boundaries, shield volcanoes	Effusive (lava flows, fountains)
Andesitic (intermediate)	52–63	Moderate	800–1000	Subduction zones, continental arcs	Mixed, often explosive stratovolcano
Rhyolitic (felsic)	>63	Very high	650–800	Continental crust, caldera systems	Violently explosive, pyroclastic-flow producing

6.2 Volcano types

- **Shield volcanoes:** broad, gently sloping ($<10^\circ$) cones built of fluid basaltic lava flows. Huge in base area but individually not very violent. Mauna Loa (Hawaii) rises 9 km from the sea floor and is the largest volcano on Earth by volume. Other examples: Kilauea, Fernandina (Galápagos), Piton de la Fournaise (Réunion).
- **Stratovolcanoes (composite):** steep-sided ($\sim 30^\circ$) cones built of alternating layers of lava and pyroclastic material. Typically andesitic, at subduction zones. Examples: Mt Fuji (Japan), Mt St Helens (USA), Mt Rainier, Vesuvius, Popocatepetl, Cotopaxi. Classic “picture-postcard” profile; high casualty potential.
- **Calderas:** large (>1 km diameter) roughly circular depressions formed when a shallow magma chamber empties rapidly and the roof collapses. May be nested inside a larger volcanic edifice. Examples: Crater Lake (Oregon, $\sim 7,700$ years ago), Santorini (Aegean, c. 1600 BCE, possibly linked to the decline of Minoan civilisation), Krakatoa (1883), Yellowstone (three “supereruptions” in the last 2.1 Ma).
- **Cinder cones:** small, steep-sided cones built of fragmented basaltic tephra. Monogenetic (erupt once). Parícutín, which appeared in a Mexican cornfield in 1943 and grew over 300 m in its first year, is the textbook example.
- **Fissure eruptions:** lava emerges along a linear fracture rather than a point vent. Laki (Iceland, 1783–84) erupted 14 km^3 of basalt along a 27 km fissure, producing a toxic sulfurous “Laki haze” that killed a quarter of Iceland’s population and up to 23,000 in Britain. The Deccan and Columbia River flood-basalt provinces are ancient mega-examples.

Volcano type comparison (to approximate scale)



6.3 Volcanic Explosivity Index (VEI)

Volcanic Explosivity Index (VEI): An 8-point logarithmic scale (Newhall & Self, 1982) classifying explosive eruption magnitude on the basis of ejecta volume, eruption plume height and qualitative descriptors. VEI 0–1 = effusive/gentle (typical Hawaiian); VEI 4 = “cataclysmic” (e.g. Galunggung 1982); VEI 6 = “colossal” (Pinatubo 1991, Krakatoa 1883); VEI 7 = “super-colossal” (Tambora 1815); VEI 8 = “mega-colossal” supereruption (Toba ~74 ka, Yellowstone ~640 ka). Each step represents roughly a 10× increase in ejecta volume.

VEI	Ejecta volume	Plume height	Descriptor	Frequency	Example
0	< 10,000 m ³	< 100 m	Hawaiian / effusive	Continuous	Kilauea lava lakes
1	> 10 ⁴ m ³	100 m–1 km	Gentle	Daily	Stromboli
2	> 10 ⁶ m ³	1–5 km	Explosive	Weekly	Galeras 1993
3	> 10 ⁷ m ³	3–15 km	Severe	Months	Nevado del Ruiz 1985
4	> 0.1 km ³	10–25 km	Cataclysmic	> 1 year	Eyjafjallajökull 2010
5	> 1 km ³	> 25 km	Paroxysmal	> 10 years	Mt St Helens 1980, Vesuvius AD 79
6	> 10 km ³	> 25 km	Colossal	> 100 years	Pinatubo 1991, Krakatoa 1883
7	> 100 km ³	> 25 km	Super-colossal	> 1,000 years	Tambora 1815 (“year without a summer”)
8	> 1,000 km ³	> 25 km	Mega-colossal (super-eruption)	> 10,000 years	Toba ~74 ka; Yellowstone ~640 ka

7. Volcanic Hazards

A single eruption can unleash a suite of hazards operating on different timescales and distances.

7.1 Lava flows

Basaltic lava flows are generally slow-moving (tens of metres per hour on gentle slopes) and rarely kill anyone in the modern era because evacuation is usually feasible. They do, however, destroy whatever they encounter: the 1973 Heimaey (Iceland) flow buried one-third of the town of Vestmannaeyjar; the 2018 lower East Rift eruption of Kīlauea destroyed >700 homes in the Leilani Estates subdivision; the 2021 Cumbre Vieja eruption on La Palma destroyed ~3,000 buildings over 85 days. Hawaiian authorities now accept lava flows as a chronic “geological tax”.

7.2 Pyroclastic flows

Pyroclastic flow (pyroclastic density current, PDC): A ground-hugging, fluidised mass of hot gas (typically 200–700 °C), volcanic ash and rock fragments travelling downslope at 80–700 km/h. The most lethal volcanic hazard; essentially unsurvivable within its footprint. Formed by column collapse, dome collapse or directed blast.

Pyroclastic flows caused the deaths at **Pompeii** (AD 79, Vesuvius) and **Saint-Pierre** (Mt Pelée, Martinique, 8 May 1902 — 28,000–30,000 dead in minutes, effectively the entire population of the town; only two survivors, including a prisoner in a stone cell). At **Mt Pinatubo 1991**, pyroclastic flows travelled up to 16 km from the vent. Accurate hazard-zone mapping can exclude settlement but the flows ignore topographic barriers below a certain height and so “safe” river valleys may funnel rather than block them.

7.3 Tephra and ashfall

Explosive eruptions eject fragmented material ranging from fine ash (<2 mm) through lapilli (2–64 mm) to bombs and blocks (>64 mm). Ashfall at distance from the vent has several impacts:

- **Respiratory hazard:** fine silica-rich ash can cause silicosis on prolonged exposure; acute respiratory distress in the immediate aftermath.
- **Roof collapse:** wet ash has a density ~1,000 kg/m³; 10 cm on a large flat roof is a substantial load. Many Pinatubo 1991 deaths came from collapsed roofs in Luzon during Typhoon Yunya, which fell on the same day.
- **Agriculture:** thin ashfall can enrich soil; thick ashfall smothers crops and pasture, poisons livestock drinking water.

- **Aviation:** volcanic ash at jet altitude scours windscreens and melts in engines, causing flame-out. The 2010 Eyjafjallajökull eruption (VEI 4) closed much of north-west European airspace for about six days, grounded ~10 million passengers and cost airlines ~\$1.7 billion — an extraordinary hazard-cascade ratio for a relatively modest eruption.

7.4 Lahars

Lahar: A volcanic mudflow or debris flow — a slurry of volcanic ash, rock and water — that follows river valleys downslope from a volcano. Can occur during an eruption (melted ice or crater-lake displacement) or years later during heavy rainfall remobilising loose tephra. Often the longest-lasting and most geographically extensive volcanic hazard.

Lahars caused the deadliest single volcanic event of the late 20th century: the **Nevado del Ruiz lahar of 13 November 1985** (Colombia) buried the town of Armero in a 5 m deep mudflow ~74 km downstream, killing 23,000 of the 28,000 inhabitants (see case study). At Pinatubo 1991, lahars recurred for years after the eruption each wet season and caused most of the total death toll.

7.5 Volcanic gases

Volcanoes emit H_2O , CO_2 , SO_2 , H_2S , HCl , HF . Sulfur dioxide is the principal short-term climatic forcing agent: if injected into the stratosphere it oxidises to sulfate aerosol and reflects incoming shortwave radiation. Pinatubo's ~20 Mt SO_2 in 1991 cooled global mean surface temperature by ~0.5 °C for ~2 years. Tambora 1815 (~60 Mt SO_2) caused 1816's "year without a summer" — failed harvests across Europe and North America, the inspiration for Mary Shelley's *Frankenstein*.

Non-eruptive gas release can also be lethal: on 21 August 1986 a cloud of CO_2 — approximately 1.6 million tonnes — burst from **Lake Nyos** (Cameroon), a volcanic crater lake, and flowed downhill as a denser-than-air mass, asphyxiating 1,746 people and thousands of livestock in villages up to 25 km away. Lake Monoun (Cameroon, 1984) and Lake Kivu (Rwanda/DRC) present similar risks.

7.6 Jökulhlaup

A sub-glacial volcanic eruption beneath an icecap melts enormous volumes of water, which builds up under the ice and releases catastrophically as a glacial outburst flood (the Icelandic term jökulhlaup is used internationally). The 1996 Gjálp eruption under Vatnajökull filled the Grímsvötn subglacial lake, which then

drained as a jökulhlaup that destroyed bridges on Iceland's ring road; peak discharge ($\sim 45,000 \text{ m}^3/\text{s}$) briefly matched the mean flow of the Congo.

7.7 Volcanic tsunami

Volcanic tsunamis are rare but possible — the 1883 Krakatoa caldera collapse generated waves up to 40 m that killed $\sim 36,000$ people around the Sunda Strait; the 2018 Anak Krakatau flank collapse without visible warning killed 437 on the coasts of Java and Sumatra. Flank collapse of volcanic islands (Canary Islands, Hawaii) is a continuing low-probability, high-impact research topic.

Exam tip — volcanic hazards form a cascade: Don't list "the hazards of a volcano" as an unordered list in the exam. Structure them by the phase of the eruption and the distance from the vent: directed blast + pyroclastic flows (proximal, minutes); ashfall and gases (regional, hours to days); lahars (along valleys, weeks); tsunami (coastal, minutes to hours); climatic cooling (global, years). This ordered approach earns higher application marks than an unordered list.

8. The Hazard Risk Framework

8.1 Risk = Hazard × Vulnerability × Exposure

Physical geoscience alone cannot explain why the same magnitude of earthquake produces vastly different death tolls in different places. The United Nations Office for Disaster Risk Reduction (UNDRR, formerly UNISDR) formalises the widely used "disaster risk" equation:

$$\text{Risk} = \text{Hazard} \times \text{Vulnerability} \times \text{Exposure}$$

Some versions divide by *Capacity* (resilience, adaptive capacity, institutional strength). Multiplication is the key point: if *any* factor is zero, risk is zero; and improvements in vulnerability or exposure can reduce risk even when the hazard itself is unchanged.

Disaster risk: The potential loss of life, injury, destroyed or damaged assets that could occur to a system, society or community in a specific period, determined probabilistically as a function of hazard, exposure, vulnerability and capacity (UNDRR 2016 terminology).

Hazard: A process, phenomenon or human activity that may cause loss of life, injury, damage to property or disruption to livelihoods. A hazard is only one element of risk — it requires exposure and vulnerability to produce a disaster.

Vulnerability: The conditions determined by physical, social, economic and environmental factors that *increase* the susceptibility of an individual, community or system to the impacts of hazards. Drivers include poverty, weak governance, poor construction, inadequate healthcare, gender inequality and social exclusion.

Exposure: The presence of people, infrastructure, housing, production capacities and other tangible human assets located in hazard-prone areas. Urbanisation in coastal or seismic zones increases exposure even if per-capita vulnerability falls.

Resilience: The ability of a system, community or society exposed to hazards to resist, absorb, accommodate, adapt to, transform and recover from the effects of a hazard in a timely and efficient manner. Includes the preservation and restoration of essential basic structures and functions.

Adaptive capacity: The ability of systems, institutions, humans and other organisms to adjust to actual or expected hazards, to moderate potential damage, to take advantage of opportunities, or to respond to consequences.

8.2 The Pressure and Release (PAR) model

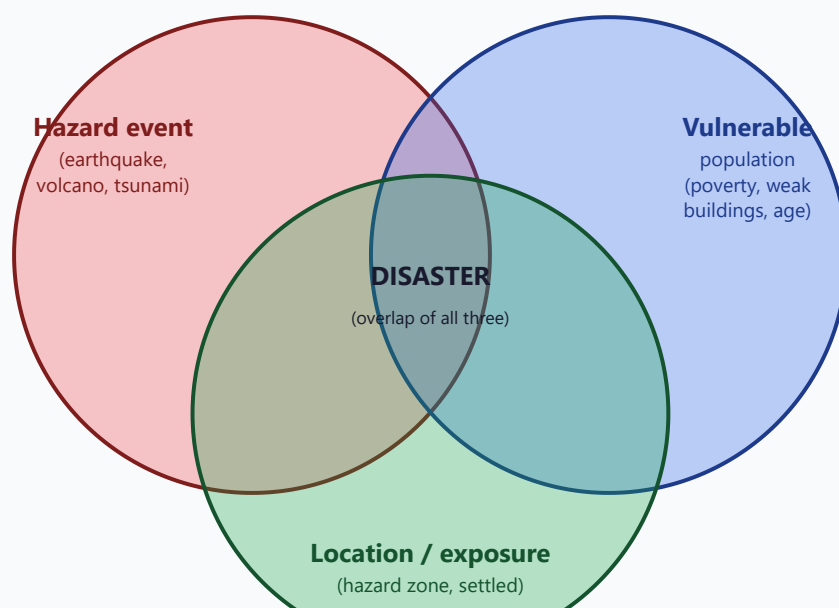
Developed by Wisner, Blaikie, Cannon and Davis (*At Risk*, 1994, 2004), the PAR model argues that disasters are the intersection of a natural hazard and a set of socially produced *unsafe conditions*. The “progression of vulnerability” runs: **root causes** (poverty, limited access to power, structures, resources) → **dynamic pressures** (rapid urbanisation, population change, conflict, environmental degradation) → **unsafe conditions** (fragile buildings on unstable slopes, insecure livelihoods, lack of preparedness). When a hazard strikes an already pressurised society, the “press” is released as disaster. The model is influential because it makes vulnerability endogenous to political economy rather than treating it as a given.

8.3 Degg’s model of hazard risk

Martin Degg's (1992) diagram shows disaster risk as the area of overlap in a Venn diagram of three circles: *hazard event*, *vulnerable population* and *human disaster*. Where all three overlap, a disaster occurs. The pedagogical value is its clarity: remove *any* of the three elements and the disaster footprint shrinks.

Degg's hazard-risk model: A Venn-style model in which "disaster" is the overlap of three sets — the hazard event, the vulnerable population, and (in some formulations) a location where these intersect. Used to emphasise that hazards alone do not cause disasters; vulnerability is equally necessary.

Degg's hazard-risk model (Venn-style)

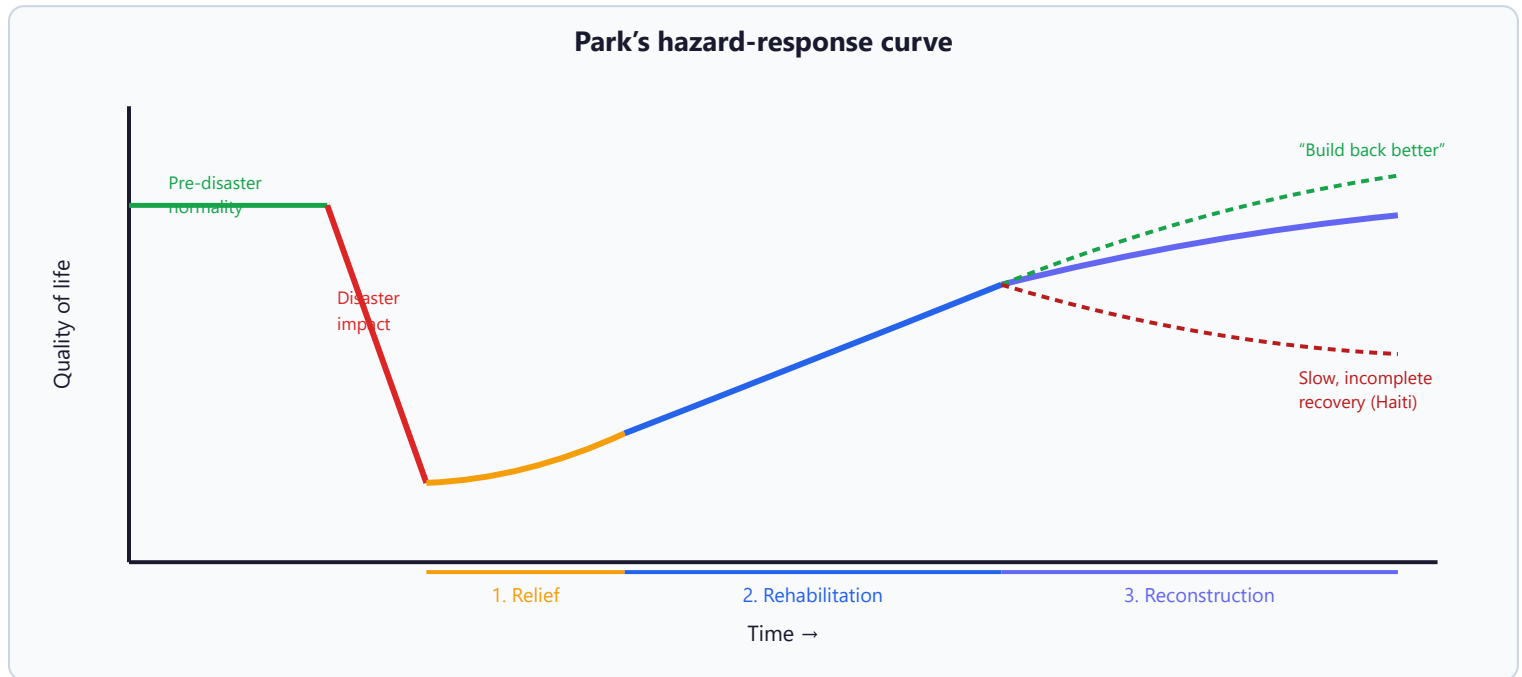


8.4 Park's hazard-response (disaster-response) curve

Chris Park's (1991) model plots *quality of life* (or equivalently social/economic functioning) on the vertical axis against *time* on the horizontal. A hazardous event causes a sudden drop, after which recovery proceeds through three overlapping phases:

1. **Relief** (hours to days) — search and rescue, emergency medical care, shelter, water and food.
2. **Rehabilitation** (days to weeks) — temporary shelter, restoration of essential services, managing displacement, beginning debris clearance.
3. **Reconstruction** (months to years) — rebuilding housing and infrastructure, "building back better", economic recovery.

Different disasters produce very different curve shapes: Haiti 2010 shows slow, incomplete recovery with a lower post-disaster baseline; Japan post-2011 shows a deeper initial drop but a faster return to near-pre-disaster functioning (though coastal Tōhoku towns are still recovering 15 years later).



Exam tip — use Degg and PAR together: Degg is excellent for a schematic introduction — “A disaster is the overlap of hazard, vulnerability and exposure”. PAR then drives the analytical depth: trace root causes, dynamic pressures and unsafe conditions. Finally apply Park’s curve to discuss *recovery*. A well-structured 38-mark essay can use all three sequentially.

9. Disaster Risk Reduction (DRR)

The Sendai Framework for Disaster Risk Reduction (2015–2030), adopted by UN member states, organises DRR around four priorities: (1) understanding risk; (2) strengthening governance; (3) investing in resilience; (4) enhancing preparedness and “build back better” recovery. In tectonic hazard contexts this translates into several tools.

9.1 Prediction and monitoring

Volcanic eruptions are now often predicted successfully at days-to-weeks lead time. Signs monitored include: *swarm seismicity* (magma forcing its way upwards); *ground deformation* (tiltmeters and InSAR satellite interferometry showing edifice inflation); *gas emissions* (SO₂ spikes measured by COSPEC or

satellites); and *thermal anomalies*. The 1991 Pinatubo evacuation, saving ~5,000 lives, is widely regarded as the showcase success of modern volcanology.

Earthquake prediction — in the sense of saying “tomorrow, at 09:00, M7, 25 km east of here” — is generally considered *not currently possible* and may be physically impossible. Claimed precursors (radon emissions, animal behaviour, electromagnetic signals) have not proven statistically reliable. What is possible is **probabilistic hazard forecasting**: identifying fault segments with long recurrence intervals (“seismic gaps”) and estimating 30-year probabilities of rupture, as the US Geological Survey’s Working Group on California Earthquake Probabilities does for California.

9.2 Early-warning systems

A distinction separate from prediction: **early-warning** means detecting an event once it has *already begun* and warning populations a few seconds to minutes ahead. Because P-waves travel faster than damaging S-waves, and electronic signals travel faster still, a few seconds of warning are sometimes possible. Examples:

- **ShakeAlert** (USGS; California, Oregon, Washington, from 2019) — delivered to mobile phones; gives a few seconds to tens of seconds of warning depending on distance from the epicentre.
- **Japan’s Earthquake Early Warning** (JMA, relayed by TV, phones and the J-Alert system); the Shinkansen’s own seismic detectors automatically halted 27 high-speed trains during Tōhoku 2011, avoiding any derailment despite the Mw 9.0 shaking.
- **Mexico’s SASMEX** (since 1991) — warnings for Mexico City of events on the Pacific subduction zone.
- **DART tsunami buoys** — NOAA network (~70 buoys) supplemented by similar systems in the Indian Ocean and Mediterranean; the Pacific Tsunami Warning Center can issue basin-wide alerts within minutes.

9.3 Building codes and seismic engineering

“Earthquakes don’t kill people, buildings do.” The deadliest mortality differential between Japan and Haiti (Section 10) is explained almost entirely by construction. Modern seismic engineering uses: *base isolation* (the building rests on rubber/lead bearings that decouple it from the ground — used in the retrofitted San Francisco City Hall and the Utah State Capitol in Salt Lake City); *tuned mass dampers* (the Taipei 101 tower’s 660-tonne pendulum); *moment-resisting steel frames*; *shear walls and reinforced concrete cores*; *cross-bracing* for unreinforced masonry retrofit. Japan, California, New Zealand (notably after the 2010–2011 Canterbury sequence) and Chile (whose 2010 Mw 8.8 earthquake killed <600 despite its enormous magnitude) enforce stringent codes.

9.4 Land-use planning and hazard mapping

Zoning out the highest-risk areas avoids placing exposure where vulnerability multiplies risk: tsunami evacuation zones on the Pacific coast of Japan (the “tsunami stones” of Iwate Prefecture carry centuries-old warnings); lahar-hazard maps around Mt Rainier with downstream evacuation routes; volcanic exclusion zones on Montserrat since the 1995 Soufrière Hills eruptions. After Tōhoku 2011, some coastal towns have been relocated to higher ground; others have built 15 m sea-walls.

9.5 Evacuation planning and drills

Japan’s annual “bosai” (disaster prevention) day on 1 September — the anniversary of the 1923 Kantō earthquake — involves nationwide drills in schools, offices, and transport networks. Schoolchildren in Kamaishi (Iwate Prefecture) practised the “tsunami tendenko” drill (each person runs for higher ground immediately, not waiting for others), widely credited with saving lives in 2011.

9.6 Education and community engagement

Cultural transmission of hazard memory is critical. The *tsunami stones* placed along the Sanriku coast after the 1896 and 1933 events (“Do not build your houses below this point”) were ignored in some villages and observed in others, with clear differential survival in 2011. Indonesian school curricula after 2004 now incorporate tsunami evacuation training; Iceland teaches every primary-school class about volcanoes and evacuation.

9.7 Insurance and financial risk transfer

In rich countries, earthquake insurance pools (California Earthquake Authority; the Earthquake Commission in New Zealand) spread losses across policyholders and reinsurers. For low-income countries, *catastrophe bonds* and sovereign risk pools (the Caribbean Catastrophe Risk Insurance Facility, CCRIF, since 2007) attempt to pre-finance recovery. The limits are well known: Haiti in 2010 had almost no domestic insurance penetration, and international aid in the hundreds of millions of dollars nonetheless failed to produce swift recovery.

9.8 International aid

Major actors include UN OCHA (Office for the Coordination of Humanitarian Affairs), the UN International Search and Rescue Advisory Group (INSARAG), national agencies (JICA Japan, USAID in the United States (dismantled in 2025, with its remaining functions moved into the State Department), DFID now FCDO United Kingdom), the Red Cross/Red Crescent movement, and NGOs such as MSF, Oxfam and CARE. Aid

can be criticised as slow, paternalistic or uncoordinated (Haiti 2010), or admired for effectiveness (the international response to the Nepal 2015 earthquake, though recovery has been slow).

10. Required Case Studies

Eduqas specifies the use of *contrasting* examples at different development levels. The following six should be known in detail; they are the principal evidence base for any Component 3 tectonic-hazards essay.

Case study 1 — Haiti earthquake, 12 January 2010 (M_w 7.0)

Tectonic setting: Enriquillo–Plantain Garden fault zone, a transform boundary between the Caribbean and North American plates; rupture at 13 km depth, ~25 km WSW of Port-au-Prince.

Impacts: Approximately 230,000 dead (Haitian government figure; independent estimates range 100,000–316,000), 300,000 injured, 1.5 million displaced; 80% of Port-au-Prince's built environment destroyed or damaged; the Presidential Palace, Cathedral and Parliament building collapsed; UN MINUSTAH headquarters collapsed, killing 102 UN staff. Economic damage ~\$8 bn (~120% of Haitian GDP).

Vulnerability drivers: Pre-existing poverty (GNI per capita ~\$660, lowest in the Western Hemisphere), post-2008 food-price crisis, political instability, weak state capacity, unregulated construction on hillsides, building codes largely unenforced, no earthquake-resistant design. Port-au-Prince had grown from 150,000 to 3 million in 60 years without planning.

Secondary cascades: Cholera epidemic from October 2010 traced genetically to Nepalese UN peacekeepers — killed a further ~10,000 and infected ~800,000 by 2016. Camp-based sexual violence. 2012 Hurricane Sandy hit a still-displaced population.

Response and recovery: Billions pledged internationally but only a fraction disbursed; the Interim Haiti Recovery Commission (co-chaired by Bill Clinton) was criticised for bypassing Haitian government; reconstruction was slow. Ten years on, many still lived in temporary shelters. Park's curve here shows a lower post-disaster baseline — incomplete recovery.

Significance: The defining modern example of how *vulnerability*, not magnitude, determines disaster outcomes.

Case study 2 — Tōhoku (Great East Japan) earthquake and tsunami, 11 March 2011 (M_w 9.0–9.1)

Tectonic setting: Mega-thrust on the Japan Trench where the Pacific Plate subducts beneath the

North American (Okhotsk) Plate. Rupture ~500 km long, ~200 km wide, focus 30 km deep, ~130 km east of Sendai. Fourth-largest earthquake ever recorded.

Primary shaking: Relatively limited structural damage for an Mw 9.0 event, testament to Japanese building codes — skyscrapers in Tokyo swayed but did not fail; Shinkansen trains stopped automatically with no derailments.

Tsunami: Ran up to 40 m on the Iwate coast (Miyako); inundated ~561 km²; overtopped sea-walls in many communities (Taro's famous 10 m "Great Wall" was overtopped). ~90% of the ~19,700 total deaths were due to the tsunami rather than the shaking.

Fukushima Daiichi nuclear disaster: The tsunami flooded the plant's diesel generators, leading to loss of cooling in reactors 1–3 and hydrogen explosions; INES Level 7 (maximum) — only the second such classification after Chernobyl 1986. ~164,000 evacuees; exclusion zone persists in parts of Fukushima Prefecture.

Economic damage: ~\$360 billion — the costliest natural disaster on record.

Recovery: Massive sea-walls up to 15 m high rebuilt along coastline; some towns (e.g. Rikuzentakata) relocated to higher ground; reconstruction ongoing into the 2020s. "Build back better" partially achieved but demographic decline in Tōhoku continues.

Significance: Demonstrates that even the most prepared country can be overwhelmed by events exceeding design thresholds; tsunami is the principal lethal hazard of subduction-zone mega-thrusts.

Case study 3 — Mt Pinatubo, Philippines, 15 June 1991 (VEI 6)

Tectonic setting: Stratovolcano on Luzon, part of the Luzon volcanic arc — the Eurasian Plate (South China Sea crust) subducts eastward beneath the Philippine Mobile Belt at the Manila Trench.

Successful prediction: Pinatubo had been dormant for ~500 years, and local populations had no memory of its hazards. In early April 1991 steam explosions began; PHIVOLCS (Philippine Institute of Volcanology and Seismology), supported by the USGS Volcano Disaster Assistance Program, installed seismometers, tiltmeters and gas-monitoring equipment. A five-level alert system was agreed with authorities. Evacuations progressed out to a 30–40 km radius by 14 June; ~60,000 people left.

Eruption: 15 June saw the climactic eruption — VEI 6, ejecting ~10 km³ of ash and tephra and ~20 Mt SO₂ into the stratosphere; plume to 34 km. Coincidence with Typhoon Yunya caused wet, heavy ashfall across Luzon.

Impacts: ~850 deaths, most from ashfall-induced roof collapse, with subsequent lahar-related deaths over the following years. Clark Air Base and Subic Bay Naval Base — major US installations — evacuated just before the eruption; the US permanently withdrew in 1992.

Lives saved: USGS and PHIVOLCS estimate that successful prediction and evacuation saved at least

5,000 lives and \$250 million in property and aircraft.

Climate effect: Global mean surface temperature dropped by $\sim 0.5^\circ\text{C}$ for ~ 2 years; stratospheric sulfate aerosol loading $\sim 3\times$ background.

Significance: The showcase example of successful volcanic prediction, joint scientific capacity-building, and cost-effective DRR. Often contrasted with Nevado del Ruiz 1985 as a success of warning and response.

Case study 4 — Nevado del Ruiz lahar, Colombia, 13 November 1985 (VEI 3)

Tectonic setting: Stratovolcano in the Northern Volcanic Zone of the Andes (Nazca Plate subducting beneath South American Plate), elevation 5,321 m with a glacier summit cap.

Eruption: Only VEI 3, relatively minor by volcanic standards. But pyroclastic flows from the summit rapidly melted the summit icecap, generating lahars that cascaded down the Río Lagunillas.

Catastrophe at Armero: The town of Armero ($\sim 28,000$ inhabitants) lay ~ 74 km from the summit. The lahars arrived at $\sim 23:00$ local time travelling at ~ 40 km/h, up to 5 m deep; $\sim 23,000$ people died within a few hours — including Omayra Sánchez, a 13-year-old girl whose three days trapped in the debris before her death on international television became a symbol of the disaster.

Warning failure: Scientists had issued hazard maps in October 1985 indicating Armero was “at high lahar risk”. Colombian authorities, distracted by ongoing political crises (the M-19 Palace of Justice siege was in early November) and reluctant to order an expensive evacuation for what they hoped would be a false alarm, did not act. Residents were told via radio that there was “no danger” during the early eruption.

Significance: A *communication and governance failure* rather than a scientific one. Shows that prediction without effective warning-response linkages saves no lives. The explicit counterpoint to Pinatubo 1991.

Case study 5 — Türkiye–Syria earthquake, 6 February 2023 (M_w 7.8 + M_w 7.5 aftershock)

Tectonic setting: East Anatolian Fault — a left-lateral transform fault where the Anatolian Plate slides past the Arabian Plate. Rupture ~ 300 km long; focal depth ~ 17 km; epicentre near Gaziantep.

Impacts: $\sim 59,000$ confirmed dead across Türkiye and northern Syria; 120,000+ injured; over 14 million affected; $\sim 300,000$ buildings destroyed or severely damaged; economic damage in Türkiye alone estimated $> \$104$ bn.

Vulnerability drivers — construction malpractice: Türkiye had modern seismic codes (2007, updated 2018) but widespread non-compliance. “İmar affı” (zoning amnesties) had legalised many

substandard buildings — over 300 contractors and officials were arrested in investigations. Many collapsed structures exhibited “soft-storey” ground floors (shops or parking on flexible pilotis) pancaking under the weight above.

Syrian side: The earthquake struck north-western Syria in territory divided between government, rebel and Kurdish controlling powers; sanctions regimes, fragmented authority and 12 years of civil war had left infrastructure, healthcare and emergency capacity in ruins. UN cross-border aid required Security Council authorisation and initially reached only Bab al-Hawa.

Recovery: >3 million internally displaced; reconstruction in Türkiye is a major state-led project; winter conditions (temperatures -5 to -10 °C) sharply increased cold-exposure mortality among survivors.

Significance: Illustrates how *political economy* (corruption, conflict, sanctions) shapes vulnerability even where technical knowledge exists. A textbook PAR-model case.

Case study 6 (secondary) — Lombok earthquake sequence, Indonesia, July–August 2018

Tectonic setting: Back-arc thrust north of the Indonesian island arc; not a mega-thrust but a shallow (~ 10 – 30 km) crustal fault. Sequence of five events M_w 6.4 to M_w 6.9 over August.

Impacts: ~ 560 dead, 1,400+ injured, $\sim 400,000$ displaced, >70,000 homes destroyed. Massive shaking-induced landslides on the slopes of Mt Rinjani trapped hundreds of hikers on the volcano, some fatally; building collapse accounted for most deaths.

Vulnerability: Lombok’s economy had pivoted sharply towards tourism (Gili Islands, Senggigi resorts); many hotel and guesthouse buildings had been erected rapidly in unreinforced masonry without seismic design. Bali’s tourism drop-off hit the Lombok economy severely.

Significance: Shows that even in middle-income, hazard-aware Indonesia, rapid economic change can outpace regulation; and that hazard impacts on tourism-dependent islands have long tails via livelihood loss.

11. Comparing Case Studies — Same Hazard, Different Outcomes

The clearest pedagogical point of this topic is that similar magnitudes produce very different mortality depending on social and economic context. Contrast Haiti 2010 and Japan 2011:

Indicator	Haiti 2010 (M_w 7.0)	Japan (Kobe) 1995 (M_w 6.9)	Japan (Tōhoku) 2011 (M_w 9.0)

Focal depth	13 km	16 km	30 km
Deaths	~230,000	~6,400	~19,700 (90% tsunami)
Displaced	1.5 million	300,000	470,000 peak
Economic damage (% GDP)	~120%	~2.5%	~4%
GNI per capita (pre-disaster, USD)	~660	~39,000	~45,000
Building code enforcement	Minimal	Strict (1981 new code)	Strictest in the world
Early warning	None	Limited (pre-ShakeAlert era)	J-Alert broadcast seconds ahead
Recovery trajectory	Slow, incomplete, lower baseline	Return to pre-disaster GDP ~5 years	Partial; coastal towns still recovering

Similarly, compare the volcanic case studies:

Indicator	Pinatubo 1991	Nevado del Ruiz 1985
VEI	6 (colossal)	3 (severe but small)
Principal hazard	Pyroclastic flows + ashfall	Lahar
Deaths	~850	~23,000
Warning issued?	Yes — PHIVOLCS + USGS, acted upon	Yes — hazard map; not acted upon
Lives saved by action	~5,000	~0 (inaction)
Key lesson	Prediction + communication + political will = success	Prediction without political action = catastrophe

Exam tip — always compare pairs: Eduqas essay questions invariably reward compared cases. Learn one pair of contrasting *earthquakes* (Haiti vs Tōhoku, or Haiti vs Chile 2010) and one pair of contrasting *volcanic events* (Pinatubo vs Nevado del Ruiz). Use them to argue the “hazards do not cause disasters — vulnerability does” thesis, weighed against counter-arguments (e.g. that Tōhoku shows even low vulnerability cannot overcome extreme hazard).

12. Super-Volcanoes and Low-Frequency High-Consequence Risk

Super-eruptions (VEI 8) are geologically rare — perhaps one per 50,000–100,000 years globally — but their consequences are civilisation-scale. Known Quaternary super-eruptions include Toba (Sumatra, ~74,000 years ago, ~2,800 km³ of ejecta; associated in some interpretations with a severe genetic bottleneck in *Homo sapiens*), the three Yellowstone cycles (2.1 Ma Huckleberry Ridge, 1.3 Ma Mesa Falls, 640 ka Lava Creek), Long Valley (California, ~760 ka), and Taupo (New Zealand, the Oruanui super-eruption ~26,500 years ago; a smaller VEI 7 eruption followed ~1,800 years ago). **Campi Flegrei** near Naples is a large, restless caldera of particular interest because it shows current unrest (slow uplift, increased seismicity) and lies beneath a metropolitan area of >3 million people.

A VEI 8 event would probably inject >1,000 Mt SO₂ into the stratosphere, producing a multi-year volcanic winter. Global food system collapse within a year is plausible: modern cereal reserves last ~70 days. Unlike nuclear war or pandemic, super-eruptions offer no plausible mitigation — only adaptive stockpiling, resilient crop varieties and international solidarity could reduce the impact. Yellowstone is intensively monitored (USGS Yellowstone Volcano Observatory, continuous GPS, seismometry, thermal imagery); current unrest is considered *not* indicative of impending eruption.

13. Climate Change and Tectonic Hazards — Emerging Research

Although plate tectonics is driven by mantle processes on timescales far longer than anthropogenic climate change, small interactions are now being investigated. Three pathways have emerging evidence:

- **Glacial isostatic adjustment (GIA):** As the Greenland and Antarctic ice sheets thin, the underlying lithosphere rebounds upwards, altering local stress fields. Iceland’s decompression of its mantle as glaciers retreat is one documented consequence — increased magma generation is modelled and may explain increased eruption frequency during past deglaciations.

- **Sea-level rise and fault-loading:** A rising ocean increases the vertical load on continental-margin faults by some fraction. Whether this is significant for rupture probability on subduction zones is debated; the effect is small relative to tectonic loading but non-zero over centuries.
- **Permafrost thaw and slope instability:** In tectonically active alpine settings (e.g. Alaska), thaw of ice-cemented rock lowers stability and could increase post-seismic landslide frequency, amplifying the secondary hazard.

These are areas of active research and should be discussed cautiously in exams — Eduqas rewards candidates who can identify that tectonic hazards are *not* immune to anthropogenic change while being clear about the modest magnitude and uncertainty of current evidence.

14. Synoptic and Evaluative Links to Other Topics

Topic 3.1 rewards cross-specification synthesis:

- **Topic 2.3 Global migration** — tectonic disasters cause displacement (Haiti 2010 diaspora to the US and DR; Syrian internal displacement after 2023).
- **Topic 2.2 Carbon and energy** — Fukushima after Tōhoku 2011 triggered Germany's *Energiewende* exit from nuclear; a tectonic event reshaped global energy politics.
- **Topic 2.4 Oceans** — subduction-zone tsunami hazard interacts with ocean-governance debates about warning systems as global public goods.
- **Topic 1.1/1.2 Landscapes** — tectonic uplift controls long-term landscape evolution; Himalayan and Alpine systems studied in Topic 1 are products of convergent tectonics.
- **Topic 2.5 21st-century challenges** — tectonic disasters test global governance and reveal inequality (Haiti vs Japan remains the synoptic exemplar).

Exam tip — evaluative essays: The Section A tectonic hazards essay is worth 38 marks and usually quotes a statement to discuss — e.g. “‘Vulnerability is more important than hazard magnitude in determining disaster outcomes.’ Discuss.” It expects a clear thesis, two pairs of contrasting case studies (Haiti/Tōhoku and Pinatubo/Ruiz), use of a framework (Degg, PAR or Park), named quantitative data (deaths, Mw, VEI, GNI per capita), and a defended conclusion. Strong answers include at least one counter-argument (e.g. Tōhoku shows magnitude can overwhelm even low vulnerability).

15. Key Vocabulary Recap

Term	Meaning in one line
Lithosphere	Rigid outer shell of Earth (crust + uppermost rigid mantle)
Asthenosphere	Ductile upper-mantle layer below lithosphere on which plates move
Slab pull	Gravitational pull of subducting dense lithosphere — dominant plate-motion driver
Focus / hypocentre	Subsurface point where rupture begins
Epicentre	Surface point directly above focus
M_w	Moment magnitude — current standard earthquake-size scale
MMI	Modified Mercalli Intensity — observed-damage scale I–XII
Benioff zone	Inclined zone of earthquakes within a subducting slab
VEI	Volcanic Explosivity Index 0–8
Pyroclastic flow	Fast, hot ground-hugging volcanic density current
Lahar	Volcanic mudflow of ash + water down valleys
Liquefaction	Loss of strength in saturated sediment during shaking
Hazard	Process that can cause harm; one element of risk
Vulnerability	Socially produced susceptibility to hazard impacts
Exposure	Presence of people/assets in hazard zones
Resilience	Ability to resist, recover from and adapt to hazards
PAR model	Pressure and Release: root causes → pressures → unsafe conditions
Degg's model	Venn diagram where disaster is the overlap of hazard, vulnerability, exposure

Park's curve	Quality-of-life vs time through relief–rehabilitation–reconstruction
DRR	Disaster Risk Reduction (Sendai Framework 2015–2030)

16. Summary — Five Things to Leave With

1. **Plate tectonics explains the distribution of hazards.** Know the boundary types and be able to locate case studies on them.
2. **Magnitude is not outcome.** Haiti vs Tōhoku shows that vulnerability — poverty, governance, construction — shapes disasters more than physical hazard magnitude does.
3. **Volcanic eruptions can be predicted; earthquakes generally cannot.** Pinatubo is the showcase; Nevado del Ruiz shows that prediction without action is useless.
4. **Use frameworks — Degg, PAR and Park.** They structure evaluative essays and signpost understanding of the hazard-vulnerability-recovery triad.
5. **DRR is a social and political project.** Building codes, land-use planning, education and early warning systems save more lives per dollar than almost any other intervention; their uneven implementation is the measure of global inequality in hazard exposure.